

but rather as a single random variable). Make distributional assumptions about the variable S and look at the volatility of S .

- Methodology 2: Recognize that 10-year CMS and 2-year CMS are each random variables, make distributional assumptions on each and use implied volatilities for each, and incorporate a “flat” (i.e., non-time varying) correlation parameter to achieve a distribution for their difference, S . Under this methodology, the correlation between 10-year CMS and 2-year CMS is assumed to remain constant through time.
- Methodology 3: Again recognize that 10-year CMS and 2-year CMS are each random variables and make distributional assumptions on each and use implied volatilities for each but now allow for a term structure of correlation (i.e., the correlation parameter varies over time).

The first methodology is not particularly robust, and it does not even consider the correlation between 10-year CMS and 2-year CMS.²⁷ The second methodology is much more general but does not allow for a time-varying correlation component, whereas the third methodology does. The difference between Methodologies 2 and 3 can be seen in Figure 9.10. Under Methodology 2, implied correlations remain constant as the expiry of the option varies. Under Methodology 3, implied correlations are assumed to change and increase as time to expiration increases.²⁸

Although Methodologies 2 and 3 are both fairly robust, the pricing of curve options using each methodology creates inconsistencies that cannot be reconciled. For example, suppose that we are pricing up a short-dated linear curve option. If we are using a flat correlation assumption as per Methodology 2, this means that we will come up with a price for the curve option that is lower than someone who is employing a time-varying correlation model as per Methodology 3.²⁹ But if we are pricing up a long-dated linear curve option, then using a flat correlation assumption as per Methodology 2 means

²⁷Note that if employing the first methodology, it would be a bad idea to assume that S follows a lognormal distribution; S can take on significantly negative values, whereas a lognormally distributed variable is always positive.

²⁸The fact that the implied correlation curve under Methodology 3 is upward sloping is related to the fact that implied volatilities for curve options typically decline as expiries increase, as shown in Figure 9.9. Recall that as implied volatilities for curve options go up, this must imply that implied correlations go down, and vice versa. Thus, since the implied volatility curve for curve options is downward sloping, that means the implied correlation curve must be upward sloping everything else the same.

²⁹Under Methodology 2, short-dated curve options are priced using a higher implied correlation than under Methodology 3. Thus, the implied volatility used under Methodology 2 must be lower, which means that the option price must be lower, assuming the option has positive vega.

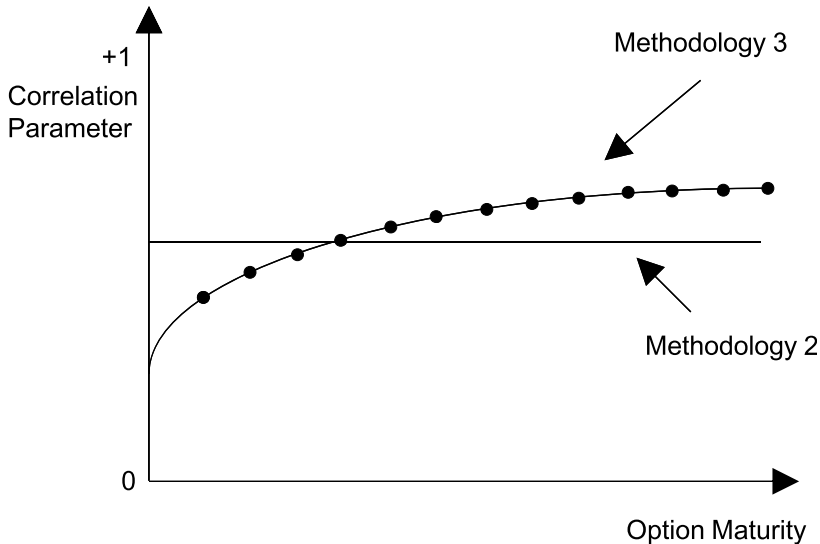


Figure 9.10: Comparison of Method 2 and Method 3

we will come up with a higher price for the option than someone using a time-varying implied correlation parameter.

As the demand for non-inversion notes grew tremendously in 2005–2006, this meant that the supply of curve options that dealers got long increased dramatically. This tended to depress the prices of curve options, which meant that the implied volatilities of curve options came down and implied correlations went up. During this time, implied correlations were trading high, certainly compared to their long-run historical averages. That is, the levels of implied correlations obtained from curve option prices were significantly higher than what was empirically observed in the market over a long period of time.

Correlation Skews

Recall that *skew* refers to the fact that options with different strikes have different implied volatilities. *Correlation skew* refers to the fact that when one computes the implied correlation from a variety of curve options with different strikes, implied correlation varies in accordance with a change in strike. For example, consider the curve caps and floors in Table 9.4 that all expire in one year.

Each of these options has a different strike and may trade at a different implied vol and therefore trade at a different implied correlation. What was

Option Type	Payoff	Strike
Curve Cap	$\max(S - 200 \text{ bps}, 0)$	+200
Curve Cap	$\max(S - 100 \text{ bps}, 0)$	+100
Curve Cap	$\max(S - 50 \text{ bps}, 0)$	+ 50
Curve Floor	$\max(0 \text{ bps} - S, 0)$	0
Curve Floor	$\max(-20 \text{ bps} - S, 0)$	-20

Table 9.4: A Variety of Curve Options and Their Strikes

typically observed in the market in 2005–2006 was that high-strike curve options traded at a lower implied correlation than did at-the-money ones. The intuition for this is as follows: high-strike curve caps come into-the-money when the curve is steep. Historically, steep yield curve environments have been associated with periods of high volatility, and vice versa. During this time we observed that implied curve vol for high-strike curve caps was high, which meant that implied correlation was low. We also observed that very low-strike curve options, which come in-the-money when the curve is significantly inverted often traded at relatively high implied vols and thus relatively low implied correlations as well.³⁰

But regardless of these perceived dynamics, it is ultimately the supply and demand of curve options of different strikes that impact the implied volatilities and therefore the implied correlations at which these options trade. As we have discussed, the popularity of non-inversion notes created a large

³⁰If an options trader sells a high-strike curve cap that is out-of-the-money today, it will have relatively low gamma. As the option approaches the money, its gamma will increase and thus the trader will be forced to delta hedge his short gamma position. For this option to approach the money means that the curve steepens out, and historically the realized volatility of the curve as well as implied vols tend to be high in a steep curve environment. Thus, the option trader will be forced to delta hedge in a situation in which he expects realized vol to be high. It is an understanding of this dynamic that, in part, has option traders selling high-strike curve options at high implied vols and therefore at low implied correlations.

Consider now an option trader who sells a very low-strike curve floor that is out-of-the-money today. It, too, will have relatively low gamma. As the option approaches the money (meaning the curve is significantly inverted), its gamma will increase and thus the trader will need to delta hedge his short gamma position. But this will happen in an inverted curve environment, and many traders believed the realized and implied vol of the curve would be high in such a curve environment. Thus, the option trader will be forced to delta hedge in a situation in which he expects realized vol to be high. Collectively, these hedging dynamics suggest that a trader will raise the implied vol that he is willing to sell both high-strike and very low-strike curve options.

supply of low-strike (binary) curve floors for dealers. While there was some interest from investors to buy these low-strike curve floors, investors were primarily interested in buying high-strike (mostly linear) curve caps, thinking that despite the flattening of the curve in 2005 and 2006, the curve would eventually re steepen. The buying of linear curve caps was a popular means of positioning for this eventual re steepening of the curve. Thus, dealers often found themselves in the position of being long low-strike binary curve floors through their selling of non-inversion notes and being short high-strike linear curve caps due to the demand for these options from investors. Thus, the forces of supply and demand created a situation in which the implied volatility of low-strike curve options traded at lower levels than for high-strike curve options. This translates into saying that low-strike curve options traded at a higher implied correlation than high-strike curve options. Very low-strike curve options (which come into-the-money when the curve is significantly inverted) also traded at relatively low implied correlations during this time. There weren't many non-inversion notes printed with very low strikes, and thus very low-strike curve options were in limited supply. These phenomena are demonstrated in Figure 9.11, which shows the *implied correlation skew* for a set of curve options with different strikes.



Figure 9.11: Typical Correlation Skew on 2s10s Curve Options with an Expiry of Two Years in 2005

9.3.6 A Couple of Trades

Let's look now at two trades involving curve options that were very popular in 2007.

Trade 1

In spring 2007 the swap curve had a rather unusual shape to it, as shown in Figure 9.12. This is what the curve looked like in the months preceding the financial crisis and reflected the debate in the market about how serious the housing crunch was going to be. Some were saying that the Fed was going to have to cut rates a lot in the coming months (and they were right), while others said that the effects of the housing crunch were overblown and would not affect the economy in any major way. As this debate raged on, the result was a swap curve that was severely inverted in the short end and was subsequently upward sloping thereafter. This unusual kink lent itself to some interesting trades that could be put together by taking advantage of the forward curve.

Using a proprietary swap model, we can derive 3-month LIBOR forwards by bootstrapping this curve and can then determine the swap curve one year

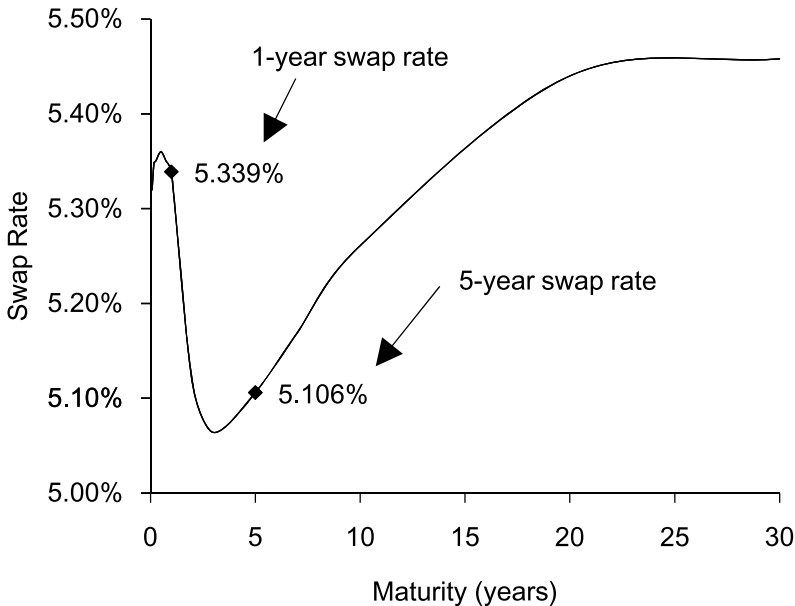


Figure 9.12: The Swap Curve in Spring 2007

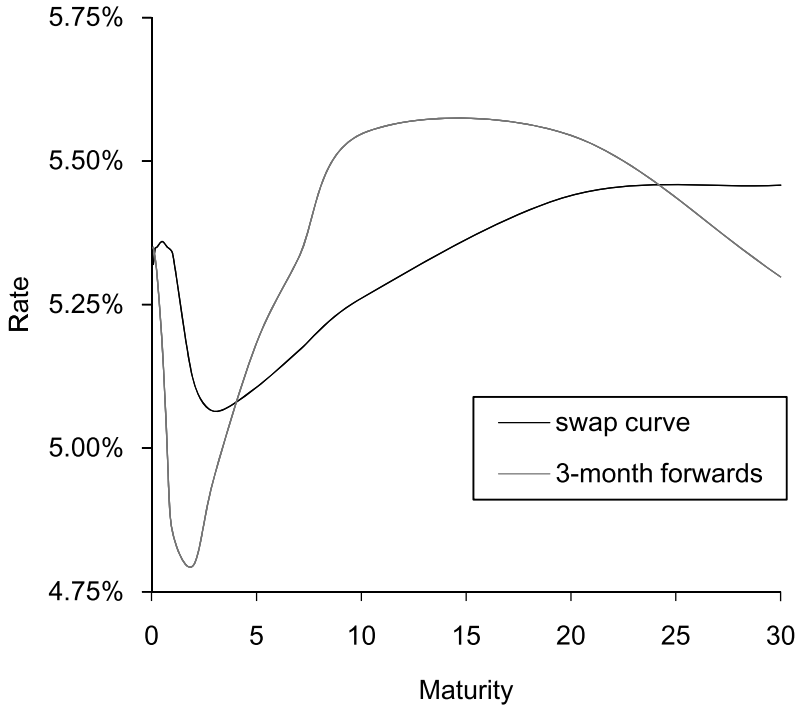


Figure 9.13: 3-Month LIBOR Forwards Derived from Bootstrapping the Curve in Figure 9.12

forward using techniques we have developed in Chapter 3. These results are shown in Figures 9.13 and 9.14. Using data from Figure 9.14, we can determine 1s5s spot and 1s5s one year forward, as displayed in Table 9.5. As indicated in this table, although spot 1s5s was inverted 23.3 bps at this time, the forwards were suggesting that 1s5s would be +20.9 bps steep in one year. This “expected” steepening of $23.3 + 20.9 = 44.2$ bps over the course of the year was a significant amount of carry. Anyone who put on steepeners would experience significant negative carry. Those who did the reverse, and put on flatteners, would experience significant positive carry.³¹ Although people were aware of the possibility of the housing crunch further deteriorating,

³¹In Chapter 8, we discussed the notion that if we put on a steepener when the curve was expected to steepen, we would earn negative carry over a holding period, whereas if we put on a flattener we would earn positive carry. Intuitively, if we want to go with the forwards, we have to pay to do so, whereas if we are willing to go against the forwards, we will be paid to do so.

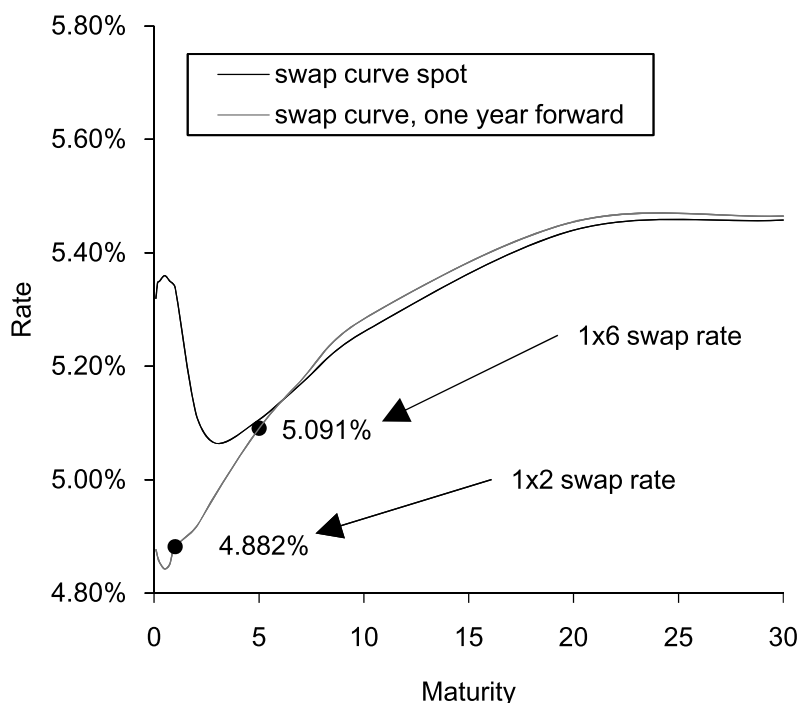


Figure 9.14: The Swap Curve in Spring 2007 and the Curve One Year Forward

	Spot	One Year Forward
5-Year Swap Rate:	5.106%	5.091%
1-Year Swap Rate:	5.339%	4.882%
1s5s:	-23.3 bps	+20.9 bps

Table 9.5: 1s5s Spot and 1s5s, One Year Forward in Spring 2007

thereby prompting the Fed to ease, they found the large positive carry from flatteners very appealing. Recognizing that they could get stopped out (i.e., hit a stop-loss) if they put on a flattener using swaps and the trade went against them, many avoided this trade.

Curve floors presented a nice opportunity for investors who were inclined to take a view against the forwards, taking the view that the curve would not

steepen out nearly as much as the 44.2 bps that the market was looking for over the next year. Consider a linear curve floor on 1s5s struck at-the-money forward that expires in one year. This floor pays out 1 bp of the notional for each bp that 1s5s is flatter than the strike at expiry. That is, the payout on this option in one year is

$$\max[\text{Strike} - (5\text{-year CMS} - 1\text{-year CMS}), 0] \times \text{Notional}. \quad (9.7)$$

What is the at-the-money forward strike on this option? It's +20.9 bps, the market's expectation of where 1s5s will be in a year.³² Suppose that at the time, this option could be purchased for 12 bps.

This was considered by many to be an attractive opportunity. This is particularly true in terms of the "carry and roll-down" on the trade. Suppose we buy this option today for 12 bps of the notional and hold the position until maturity. Assuming an unchanged spot swap curve, the payout from this option would be

$$\max[+20.9 \text{ bps} - (5.106\% - 5.339\%), 0] = 44.2 \text{ bps}, \quad (9.8)$$

for a net profit (excluding NPV considerations) of $44.2 \text{ bps} - 12 \text{ bps} = 32.2 \text{ bps}$, a nice trade.

What actually happened? What actually happened was not nearly as pretty as this carry and roll-down analysis. Over the course of the next year, the financial crisis worsened and the Fed cut short-term rates and the curve steepened out significantly. At the time of expiration, the swap curve was as depicted in Figure 9.15. As this figure shows, 1s5s was $3.283\% - 2.362\% = +92.1 \text{ bps}$ at expiration. Thus, the option finished very far out-of-the-money. So an investor who bought this option would have lost the premium paid at trade inception and would have received no payout at expiration.

Although this trade did not perform well, it was in many ways a much safer trade than if someone had put on a 1s5s flattener outright. For example, suppose an investor wanted to risk \$100,000 per bp on the flattener. That is, the investor was interested in putting on a trade in which every basis point change in 1s5s results in a payout change of \$100,000. If the investor bought the curve floor, this means that the investor would buy a notional of \$1bn. The premium of 12 bps of the notional would have been paid with no subsequent payout from the option, amounting to a loss of \$1.2mm. Alternatively, suppose the investor put on an outright forward flattener by paying

³²Technically, the at-the-money forward strike on this option is not +20.9 bps. That differential was computed using the difference in the two relevant forward swap rates. As we discussed in Chapter 6, there is a convexity adjustment that must be made to swap rates that are being paid out as underlying variables. Once again, we will ignore such adjustments for our purposes throughout this chapter.

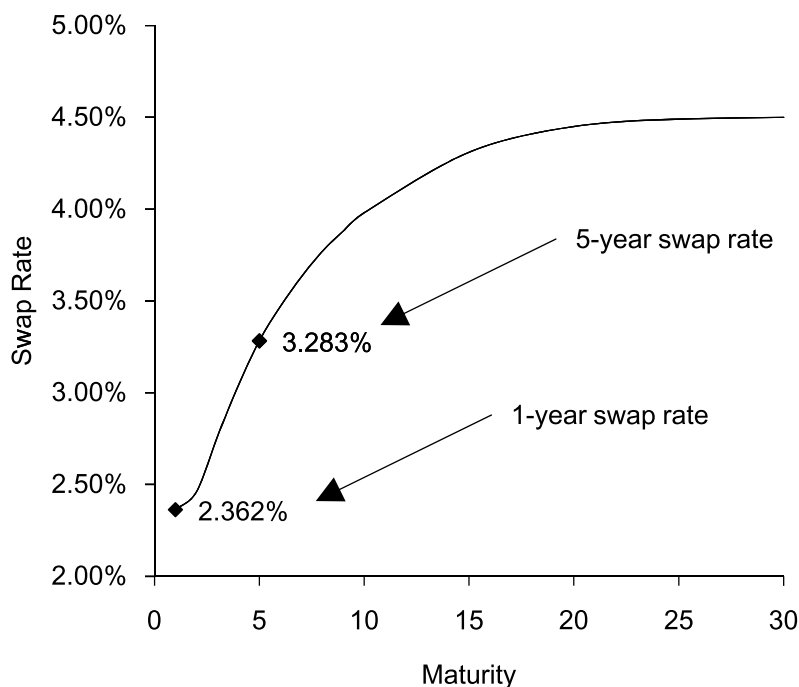


Figure 9.15: The Swap Curve in Spring 2008

on a 1x2 and receiving on a 1x6 and chose notional such that he was risking \$100,000 per bp. This trade would have been put on at a rate differential of +20.9 bps, the difference between the two forward starting swap rates. Assuming this trade was held for a year, 1s5s finishes at +92.1 bps, resulting in a loss of $+92.1 - 20.9 = 71.2$ bps, or approximately \$7.12mm.³³

Trade 2

The following trade opportunity was available in 2007 and exploited the shape of both the forward curve and the vol curve for curve options. Refer now to Figure 9.16, the top half of which shows the vol curve for curve

³³Most investors would have probably hit a predetermined stop-loss on the trade prior to one year going by, given the large move in the curve, and unwound the trade somewhere along the way. The nice thing about the version of the trade using the curve option is that it gives one a well-defined downside equal to the premium paid for the option.

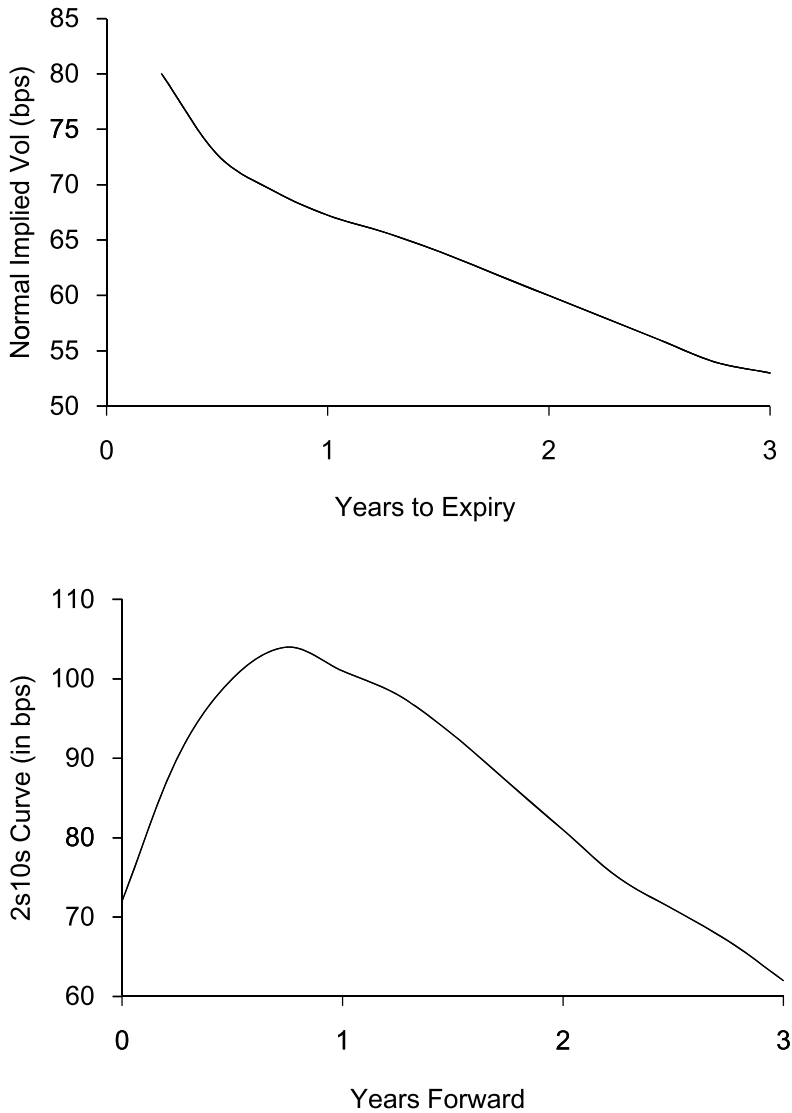


Figure 9.16: The Vol Curve for Curve Options (*top half*) and the 2s10s Curve, Spot and Forward (*bottom half*)

options. As we have discussed previously, short-dated curve options typically trade at a higher implied vol than do long-dated options, and this differential is rather pronounced. The bottom half of this figure shows 2s10s, both spot and forward, at this time. For example, 2s10s spot was around 73